

NEW HORIZON COLLEGE OF ENGINEERING, BANGALORE

Department of Electronics and Communication Engineering.

INTRODUCTION TO ELECTRONICS AND COMMUNICATION ENGINEERING – 25ESC231 (QUESTION BANK)

Q. NO	MODULE 1
1	Explain the V-I characteristics of a PN junction diode.
2	Explain the following concerning the semiconductor diode. PIV, Reverse saturation current, threshold voltage, breakdown voltage, Forward voltage drop.
3	Illustrate the working of a half-wave rectifier with a neat circuit diagram and waveform.
4	Illustrate the working of a full wave bridge rectifier with a neat circuit diagram and waveform.
5	Illustrate the working of a full wave center tapped rectifier with a neat circuit diagram and waveform.
6	Define Biasing. Explain forward and reverse biasing of diode.
7	Explain the working of a full wave bridge rectifier with filter and voltage regulator with a neat circuit diagram and waveform.
8	Derive the expression of efficiency and ripple factor for a full wave and half wave rectifier.
9	Explain the working of a Centre tapped full wave rectifier with filter and voltage regulator with a neat circuit diagram and waveform.

Q. NO	MODULE 2
1	Explain the construction and operation of BJT.
2	Explain how BJT transistor can be used as switch and amplifier with necessary circuits.
3	Explain the construction and operation of JFET.

4	Explain the working of a single stage Common Emitter amplifier.
5	Explain the DC load line of a CE configuration and discuss the significance of the Q point.
6	Explain the Input and Output characteristics of BJT transistor.
7	Compare BJT and JFET.

Q. NO	MODULE 3
1	List and define any five parameters of Op-Amp.
2	Explain and derive the gain of Inverting Amplifier with necessary equations and circuit diagram.
3	Explain and derive the gain of Non Inverting Amplifier with necessary equations and circuit diagram.
4	Explain the working of Wein Bridge Oscillator with necessary circuit diagram and equations.
5	Explain the working of RC Phase shift Oscillator with necessary circuit diagram and equations.
6	Explain the need for oscillations.

Q. NO	MODULE 4
1	What is modulation and Explain the need for Modulation.
2	Explain the elements of the Communication System
3	Explain super heterodyne receiver.
4	Explain AM, FM, and PM modulation Techniques with necessary waveforms.
5	Explain ASK, FSK, and PSK modulation Techniques with necessary waveforms.
6	List out the differences between analog and digital communication systems.

7	List the advantages and disadvantages of AM, FM and PM.
8	List the advantages and disadvantages of ASK, FSK and PSK.

Q. NO	MODULE 5
1	Explain any five logic gates with a truth table and logic symbol.
2	Explain and prove De-Morgan's theorem with 2 and 3 variables.
3	Prove: (a) Associative Law (b) Distributive Law
4	Implement NOT, AND & OR gate using only NAND gates and NOR gates.
5	Solve the following using 1's and 2's complement 1. $(33)_{10} - (18)_{10}$ 2. $(18)_{10} - (33)_{10}$
6	Subtract $(10110)_2 - (1110)_2$ using 1's and 2's complement method. Subtract $(10001)_2 - (1001)_2$ using 1's and 2's complement method.
7	Simplify the following Boolean expression i) $f(x, y, z) = x'y' + yz + x'yz'$ ii) $f(a, b, c) = ab + a'b'c' + a'bc'$ Refer class notes for more problems
8	Convert: i) $(ABCD.12)_{16} = (?)_8 = (?)_2$ ii) $(726.55)_{10} = (?)_2 = (?)_8$ Refer class notes for more problems